Sequelae of Eating Disorders at Imaging

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Abbreviation: ED = eating disorder

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Describe the clinical presentation of patients with a variety of EDs.
- Discuss the wide range of medical complications and sequelae of EDs.
- Identify the imaging features associated with common and uncommon sequelae of EDs and discuss management options.

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Although eating disorders are common, they tend to be underdiagnosed and undertreated because social stigma tends to make patients less likely to seek medical attention and less compliant with medical treatment. Diagnosis is crucial because these disorders can affect any organ system and are associated with the highest mortality rate of any psychiatric disorder. Because of this, imaging findings, when recognized, can be vital to the diagnosis and management of eating disorders and their related complications. The authors familiarize the radiologist with the pathophysiology and sequelae of eating disorders and provide an overview of the related imaging findings. Some imaging findings associated with eating disorders are nonspecific, and others are subtle. The presence of these findings should alert the radiologist to correlate them with the patient's medical history and laboratory results and the clinical team's findings at the physical examination. The combination of these findings may suggest a diagnosis that might otherwise be missed. Topics addressed include *(a)* the pathophysiology of eating disorders, *(b)* the clinical presentation of patients with eating disorders and their medical complications and sequelae, *(c)* the imaging features associated with common and uncommon sequelae of eating disorders, *(d)* an overview of management and treatment of eating disorders, and *(e)* conditions that can mimic eating disorders (eg, substance abuse, medically induced eating disorders, and malnourishment in patients with cancer).

Online supplemental material is available for this article.

Introduction

The term *eating disorders* (EDs) describes a range of psychiatric disorders characterized by "persistent disturbance of eating or eating-related behavior that results in the altered consumption or absorption of food and that significantly impairs physical health or psychosocial functioning" (1).

The prevalence of EDs is 0.3%–1.0% of the population, with an incidence of eight cases per 100 000 people per year and a female predominance (2). In men, the prevalence is approximately 10 times lower (2). The causes of EDs are multifactorial and include genetic predispositions, personality traits, anxiety disorders, family history of depression or obesity, and family or cultural pressure pertaining to appearance (2).

TEACHING POINTS

- EDs are a substantial health threat because they can affect any organ system and are associated with the highest mortality rates of any psychiatric disorder.
- Subacute combined degeneration results from degeneration of the dorsal and lateral columns of the spinal cord secondary to vitamin B12 deficiency. Although this is typically associated with pernicious anemia, it can occur in patients with a restricting ED. CT and MRI demonstrate mild cord enlargement or swelling. T2-weighted MRI shows symmetric abnormally high signal intensity of the dorsal and lateral columns, mainly in the cervical and thoracic cord. Mild enhancement of the dorsal columns may be seen at gadolinium-enhanced T1-weighted MRI.
- Small heart syndrome is characterized by the presence of a small heart shadow (ie, a cardiothoracic ratio ≤ 0.42) on a posteroanterior chest radiograph and is associated with nonspecific clinical symptoms including weakness, fatigue with mild exertion, and dyspnea. Although other medical conditions such as cancer-induced cachexia can manifest with small heart syndrome, the presence of this finding in young patients with no underlying health issues should raise suspicion for underlying EDs.
- Repetitive vomiting is associated with increased laxity of the lower esophageal sphincter, which may manifest as a patulous gastroesophageal junction at fluoroscopy and can even result in esophageal rupture (ie, Boerhaave syndrome) if vomiting is aggressive. Boerhaave syndrome is associated with a high mortality rate without early diagnosis and surgical repair, and it can be indicated by the presence of pneumomediastinal or pleural effusions at chest radiography or chest CT, mediastinal fluid or inflammation at chest CT, or a leak at fluoroscopy.
- Disorders associated with binging, including bulimia nervosa, can result in chronic distention of the stomach, with an increase in stomach capacity to over 15 L. Normally, rupture can occur with intragastric pressures greater than 120 mm Hg (approximately 4 L of fluid). Binging episodes can lead to acute gastric dilatation, which can have potentially life-threatening complications including gastric necrosis, perforation or rupture, and even abdominal compartment syndrome.

Patients with EDs commonly demonstrate anxiety, depression, and obsessive tendencies (3). Serotonin pathways are implicated in modulation of mood, impulse regulation, and obsessiveness (3). Studies of patients with active anorexia nervosa and bulimia show reduced serotonergic activity during the acute phase of the illness, and multiple SPECT and PET radioligand binding studies have shown the persistence of altered activity in the serotonin pathway even after recovery (3).

EDs are a substantial health threat because they can affect any organ system and are associated with the highest mortality rates of any psychiatric disorder. Anorexia nervosa is associated with a 5%–10% mortality rate within 10 years of diagnosis (2,4). Despite their commonness and association with high morbidity and mortality rates, EDs tend to be underdiagnosed and undertreated because social stigmas make patients less likely to seek medical attention and less compliant with treatment (4).

The most well-recognized EDs are anorexia nervosa and bulimia nervosa, but multiple types and subtypes of EDs have been described (Table 1). Most behaviors that characterize EDs can be generally categorized as *(a)* restricting, *(b)* purging, or *(c)* binging and other forms of disordered ingestion. Manifestations of restricting disorders mainly result from complications of malnutrition due to a combination of nutrient deficiencies, dehydration, electrolyte abnormalities, metabolic alterations, and hormonal dysregulation. Although a few sequelae of repetitive purging behaviors are associated with electrolyte abnormalities, hormonal dysregulation, and even nervous system dysregulation, most manifestations occur directly (caused by repetitive physical insult from the behavior itself) or indirectly (as an adaptation to prolonged irritation). Repetitive physical insult or irritation and subsequent adaptation are long-term manifestations of EDs, but many sequelae of binging and disordered ingestion tend to be more acute, causing trauma to the organs involved. Because so many physiologic processes are disrupted, the manifestations are multisystemic and widely varied (Table 2).

Treatment of EDs is generally a combination of medical and nutritional management (eg, correcting electrolyte abnormalities and vitamin deficiencies and adjusting and monitoring the patient's diet) and psychologic and psychiatric therapy (eg, medications, behavioral therapy, and electroconvulsive therapy), although some complications are severe enough to warrant surgical intervention. A particularly serious complication of anorexia nervosa is refeeding syndrome. During prolonged starvation, major hormonal and metabolic changes take place and intracellular minerals become severely depleted (although serum concentrations may remain deceptively normal) (5). During refeeding, normal biochemical processes result in the intracellular absorption of various substances (glucose, electrolytes), leading to depletion in the serum levels of phosphate, potassium, and magnesium. The clinical features of refeeding syndrome result from the functional deficits associated with these electrolyte deficiencies and the rapid change in basal metabolic rate (5).

Imaging findings may suggest the diagnosis of EDs and are important for identifying complications (Table 3). The aims of this article are to *(a)* familiarize the radiologist with the pathophysiology of EDs, *(b)* discuss the clinical presentation of patients with EDs and their medical complications and sequelae, *(c)* describe the imaging features associated with common and uncommon

Table 1: Spectrum of EDs and Their Characteristics

Eating Disorder	Characteristics
Anorexia nervosa	Food restriction leading to significantly low body weight Fear of gaining weight or persistent behaviors preventing weight gain Distorted body image, overemphasis on weight or shape as determinants of self-worth, or denial of medical problems associated with low body weight
Bulimia nervosa	Recurrent episodes of binge eating Compensatory behaviors (eg, induced vomiting; use of laxatives, diuretics, or enemas; fasting; or excessive exercise) that occur at least once per week for 3 months Overemphasis on weight or shape as determinants of self-worth
Binge eating disorder	Three of the following: More rapid eating than normal Eating until uncomfortably full Eating large amounts of food even in the absence of hunger Eating alone because of embarrassment about the amount of food consumed Feelings of disgust, depression, or guilt after binging No regular compensatory behaviors
Purging disorder	Recurrent purging episodes in absence of binging
Night eating syndrome	Recurrent episodes of eating after awaking from sleep, or eating excessively after an evening meal
Rumination disorder	Repeated regurgitation of food Food may be rechewed, reswallowed, or spit out
Pica	Repeat consumption of nonnutritional nonfood substances Eating habits inappropriate for stage of development and not culturally or socially normal

Source.—Reference 1.

sequelae and complications, and *(d)* provide an overview of management and treatment of EDs.

Multisystemic Radiologic Manifestations of Multiple and Mixed EDs

Cardiac Manifestations

Cardiac complications develop in up to 80% of patients with EDs because of a combination of malnutrition, electrolyte imbalances, and depletion of intravascular volume (4). Higher incidences of bradycardia, hypotension, mitral valve prolapse, decreased left ventricular mass (secondary to muscular atrophy), lower cardiac output, and lower cardiac indexes are observed In patients with anorexia (4). The myocardial atrophy frequently seen at pathologic examination leads to abnormalities of conduction and repolarization that predispose patients to arrhythmia and sudden cardiac death, which are common in these patients (4).

Urologic Manifestations

Vomiting and abuse of laxatives or diuretics can lead to volume depletion (ie, reduction in the volume of extracellular fluid), to which the kidneys respond by increasing the secretion of aldosterone, which subsequently increases the absorption of sodium and bicarbonate and the excretion of potassium (6). This results in electrolyte disturbances and problems with water retention (2). Acute and chronic kidney failure can occur (2). Nephrolithiasis is seen in 5% of patients with anorexia nervosa (Fig E1) (2). Calculi of ammonium urate form because of the low output of urine and increased output of urinary ammonium from hypophosphaturia and diarrhea (2). In addition, volume depletion promotes urinary saturation with uric acid, oxalate, and calcium and can result in formation of uric acid and calcium oxalate kidney stones, which may lead to nephrocalcinosis (2).

Gastrointestinal Manifestations

Both restrictive and purging disorders are associated with gastric dysmotility and prolonged colonic transit time (Fig E2) (7). General gastrointestinal discomfort is common in patients with EDs, and they may present with irritable bowel syndrome, constipation, and even incontinence (Fig E3) (8). Constipation in patients with EDs occurs because of a combination of delayed gastrointestinal transit, the physical changes of chronic malnutrition, and the dysfunction of the pelvic floor due to muscle damage from repetitive vomiting and straining (8). Severe complications of gastrointestinal dysmotility include fecal impaction in the colon, obstruction, and necrotizing colitis (Fig 1) (8). Although necrotizing colitis is rare, it has been observed in patients with severe malnutrition and during refeeding and is thought to be caused by starvation-induced hypoperfusion, hypoxic-ischemic injury, dysmotility, and mucosal injury (8).

Cases of pancreatitis have been reported in patients with anorexia and bulimia (9). A number of mechanisms have been proposed to explain the development of pancreatitis in patients with EDs, among which are microlithiasis, ischemia, and structural damage (Fig 2) (8). Acute pancreatitis is rare but tends to be associated with severe malnutrition, typically developing during refeeding because of shifts in fluids (8). Duodenal dysmotility, which results in retrograde flow of the contents of the duodenum, has also been implicated (8).

Restricting Disorders

Manifestations due to Nutritional Deficiencies

Neurologic Manifestations.—Brain volume loss occurs because of severe malnutrition and deficiencies in fatty acids, proteins, and neuroprotective nutrients (B-complex vitamins and antioxidants) (10). A deficiency of insulin growth factor-1 (IGF-1) is theorized to decrease oligodendrocyte proliferation and differentiation, which inhibits myelination and ultimately leads to a reduction in brain volume (10). In adolescents, severe malnutrition results in modification in the maturation of the brain and cortical architecture to save energy and preserve network efficiency. Dehydration also results in alterations in osmosis that shift fluid to the extracellular space (10).

Findings of volume loss are more pronounced in patients with anorexia nervosa than in those with bulimia (11). CT of the head shows a generalized loss of volume, ventricular enlargement, and enlargement of the cerebral spinal fluid spaces (Fig 3) (12). MRI shows diffuse loss of gray matter, enlarged cerebral spinal fluid spaces and ventricles, and widened sylvian fissures (5). Some studies (10,11) have demonstrated an improvement in the volume of gray and white matter after the patient gains weight. MRI is useful for identifying additional regions that are more susceptible to the loss of volume, including the parietal cortex, precuneus insula, thalamus, and limbic structures, including the cingulate cortex, hippocampus, and amygdala (5). The pituitary gland is also thought to atrophy as a result of chronic malnutrition (8). Diffusion-tensor MRI shows alterations in the microarchitecture of the white matter, mainly in thalamocortical and interhemispheric connections, tracts connecting different cortical regions, and tracts in the lim-

Table 2: Manifestations of EDs Stratified by Pathogenesis

Dehydration and electrolyte abnormalities
Central pontine myelinolysis
Cardiac arrhythmia
Nephrocalcinosis or nephrolithiasis
Pancreatitis
Renal failure
Metabolic alterations and changes in tissue composition and distribution
Bone marrow adipose tissue
Enophthalmos
Fat reduction and redistribution
Hepatic steatosis
Pancreatic atrophy
Patulous eustachian tube
Superior mesenteric artery syndrome
Nutrient deficiencies
Ascites, delayed skeletal growth, volume loss
Emphysema, pneumothorax, pneumomediastinum
Pericardial effusion
Pancreatitis
Small heart syndrome or Takotsubo cardiomyopathy
Subacute combined degeneration or vitamin C deficiency (scurvy)
Wernicke encephalopathy
Hormonal dysregulation
Female genital tract immaturity
Osteopenia or osteoporosis
Peripheral edema
Pericardial effusion
Nervous system dysregulation
Cathartic colon, constipation, or delayed gastric emptying
Pancreatitis
Parotid gland swelling
Repetitive physical insult, prolonged irritation
Dental caries
Aspiration, infection, abscess, acute respiratory distress syndrome, or lipoid pneumonia
Barrett esophagus or patulous gastroesophageal junction
Boerhaave syndrome
Esophageal carcinoma
Esophageal strictures, esophageal rupture, esophagitis
Gastric wall thickening, gastritis, cathartic colon, ulcers, or constipation
Disordered ingestion
Abdominal compartment syndrome
Acute gastric dilatation, gastric necrosis, perforation, or rupture
Superior mesenteric artery syndrome
Foreign bodies or bezoars
Bowel obstruction or perforation

Table 3: Imaging Manifestations of EDs

Type of Manifestation	Manifestations According to ED Behavior: Restricting	Purging (Vomiting or Laxative Abuse)	Binging and Other Disorders (Pica)
Cardiac	Small heart syndrome Mitral valve prolapse Pericardial effusion Takotsubo cardiomyopathy	. . .	. . .
Gastrointestinal	Esophagitis, gastritis, or ulcer Gastric or duodenal dysmotility Bowel obstruction or perforation Hepatic steatosis Pancreatic atrophy or pancreatitis Prolonged colonic transit time or constipation Colonic impaction or obstruction Necrotizing colitis Superior mesenteric artery syndrome	Esophagitis, gastritis, ulcer Esophageal strictures Patulous gastroesophageal junction Barrett esophagus Esophageal carcinoma Boerhaave syndrome or esophageal rupture Pancreatitis Gastric or duodenal dysmotility Prolonged colonic transit time or constipation Cathartic colon or melanosis coli	Acute gastric dilatation Gastric necrosis, perforation, or rupture Abdominal compartment syndrome Superior mesenteric artery syndrome Esophagitis, gastritis, or ulcers Gastric or duodenal dysmotility Prolonged colonic transit time or constipation Bowel obstruction or perforation Foreign bodies and bezoars (pica)
Head and neck	Enophthalmos Patulous eustachian tube	Dental caries Parotid gland swelling or sialadenosis	. . .
Musculoskeletal	Osteopenia or osteoporosis Insufficiency or fragility fractures Marrow atrophy Delayed skeletal growth Vitamin C deficiency (scurvy) Osteopenia and osteoporosis	. . .	. . .
Neurologic	Volume loss Central pontine myelinolysis Wernicke encephalopathy Subacute combined degeneration	Volume loss	. . .
Pulmonary	Emphysema Pneumothorax Pneumomediastinum	Aspiration, infection, or abscess Lipoid pneumonia	. . .
Soft tissue	Deficiency of subcutaneous fat High fat attenuation	. . .	. . .
Urogenital	Nephrolithiasis Nephrocalcinosis Female genital tract immaturity Peripheral edema	Nephrolithiasis Nephrocalcinosis	. . .

bic system. Thus far, the literature (10) shows that white matter is affected mainly in young patients with anorexia. Loss of volume can be complicated by a subdural hematoma in severe cases (13).

Wernicke encephalopathy is a neurologic condition that is commonly associated with a deficiency of vitamin B1 (thiamine) secondary to chronic use of alcohol (14). It can also be precipitated by disorders that result in severe malnutrition, such as anorexia nervosa (with thiamine deficiency seen in 38% of these patients) (14). The classic clinical presentation for these patients includes altered mental status, ataxia, and ophthalmoplegia, although patients with nonalcoholic Wernicke encephalopathy tend to present atypically (14). CT findings are often negative for abnormalities in the acute phase but can show areas of reduced attenuation in the periaqueductal gray matter and medial aspects of the thalami (Fig E4) (14). MRI has

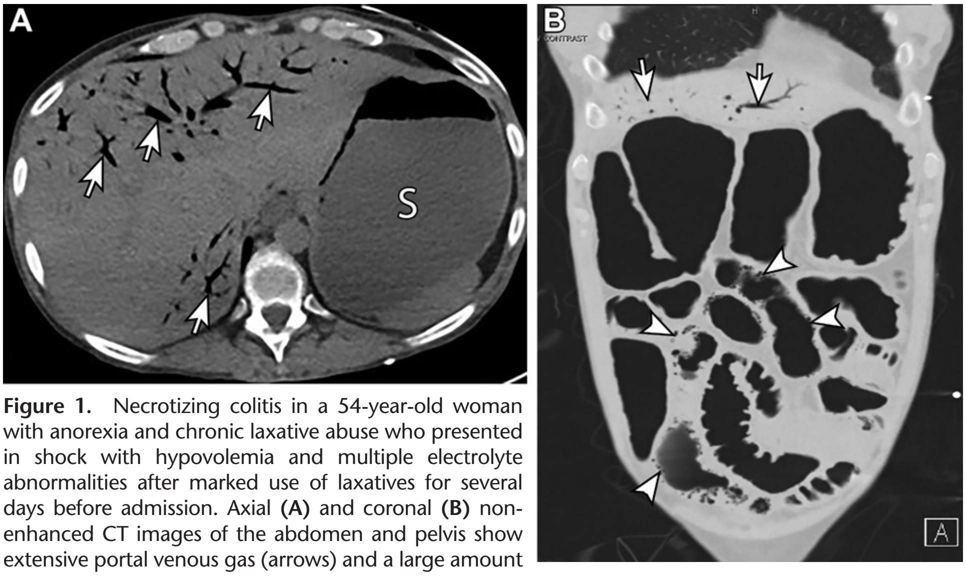


Figure 1. Necrotizing colitis in a 54-year-old woman with anorexia and chronic laxative abuse who presented in shock with hypovolemia and multiple electrolyte abnormalities after marked use of laxatives for several days before admission. Axial **(A)** and coronal **(B)** nonenhanced CT images of the abdomen and pelvis show extensive portal venous gas (arrows) and a large amount of pneumatosis throughout the visualized bowel (arrowheads in **B**). There is associated diffuse bowel ileus or paralysis. Note the distended stomach *(S)* and the paucity of subcutaneous fat.

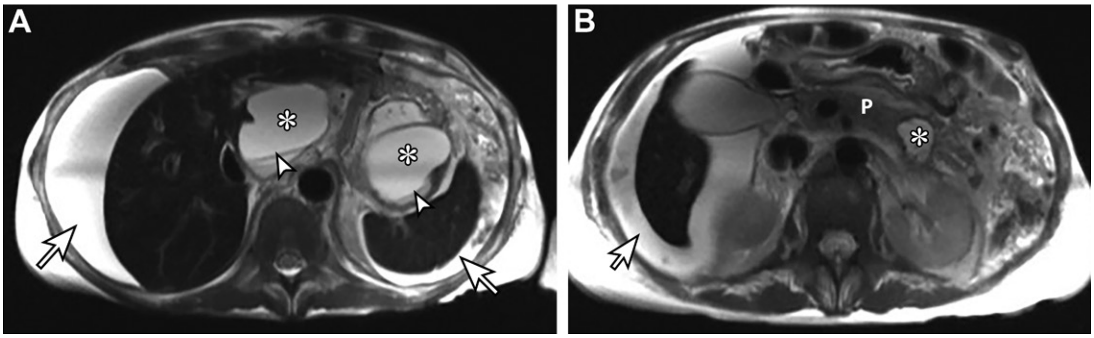
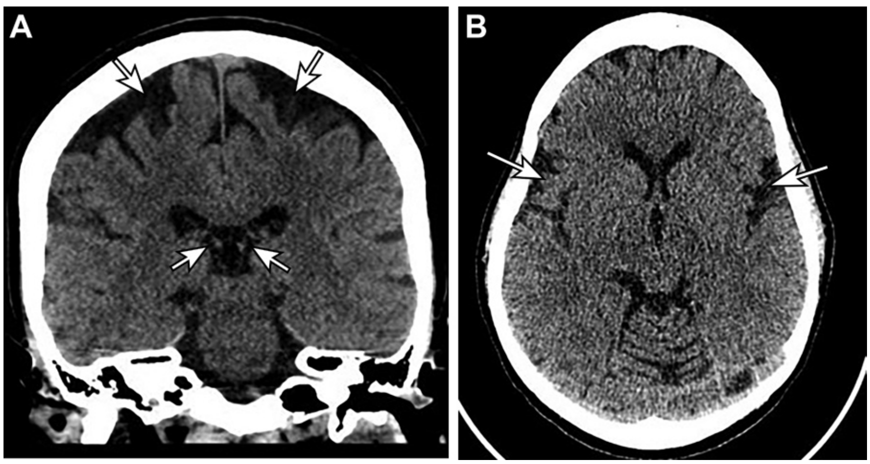


Figure 2. Acute pancreatitis in a 47-year-old woman with anorexia. Axial T2-weighted images of the abdomen reveal multiple peripancreatic and intrapancreatic fluid collections (*) with fluid-fluid levels (arrowheads in **A**) indicative of debris rather than blood products. Moderate ascites is also seen (arrows). *P* = pancreas.

Figure 3. Volume loss in a 41-year-old woman with anorexia nervosa. Nonenhanced coronal **(A)** and axial **(B)** CT images of the brain show ventricular and sulcal enlargement, with cerebrospinal fluid spaces (arrows in **A**) larger than expected for a patient of this age. Note the widening of the sylvian fissures (arrows in **B**).



low sensitivity (53%) but high specificity (93%) for diagnosis of Wernicke encephalopathy (14). Focal abnormalities tend to be bilateral and symmetric, demonstrating increased signal intensity at T2-weighted MRI, fluid-attenuated inversion-recovery MRI, and diffusion-weighted MRI and may enhance at contrast-enhanced T1-weighted MRI (in approximately 50% of cases) (Fig E4) (14). Although certain regions and structures tend to be affected more commonly, unusual sites

Table 4: Regions of Brain Affected by Wernicke Encephalopathy Detectable at MRI

Typical locations	
	Hypothalamus
	Mamillary bodies
	Medial thalamus
	Periaqueductal gray matter
	Tectal plate
Atypical locations	
	Caudate putamen
	Cerebellar vermis
	Cranial nerve nuclei
	Dentate nuclei
	Dorsal medulla
	Fornix
	Paravermian region
	Pons
	Postcentral gyrus
	Precentral gyrus
	Red nucleus
	Splenium of corpus callosum
	Substantia nigra

Source.—Reference 14.

that are documented in the literature (14) can be affected in conjunction with more typical locations (Table 4) .

Subacute combined degeneration results from degeneration of the dorsal and lateral columns of the spinal cord secondary to vitamin B12 deficiency (15). Although this is typically associated with pernicious anemia, it can occur in patients with a restricting ED (15). CT and MRI demonstrate mild cord enlargement or swelling (16). T2-weighted MRI shows symmetric abnormally high signal intensity of the dorsal and lateral columns, mainly in the cervical and thoracic cord (Fig E5) (15,16). Mild enhancement of the dorsal columns may be seen at gadolinium-enhanced T1-weighted MRI (15,16).

Cardiac Manifestations.—Generally observed in particularly malnourished patients, small heart syndrome is characterized by the presence of a small heart shadow (ie, a cardiothoracic ratio ≤ 0.42) on a posteroanterior chest radiograph and is associated with nonspecific clinical symptoms including weakness, fatigue with mild exertion, and dyspnea (Fig 4) (17,18). Although other medical conditions such as cancer-induced cachexia can manifest with small heart syndrome, the presence of this finding in young patients with no underlying health issues should raise suspicion for an underlying ED.

Mitral valve prolapse is reported in up to 83% of patients with EDs and is believed to result from reduction in the size of the left ventricle (7). This finding may be seen in real time at echocardiography or may be viewed with cardiac MRI. A dilated left atrium at chest radiography or CT may also suggest the diagnosis. Pericardial effusion is seen in approximately 35% of patients with anorexia nervosa, occurring because of a combination of protein deficiency and low levels of thyroid hormones (19). Although pericardial effusion is often small and without hemodynamic significance, severe cases of pericardial effusion in which cardiac tamponade required intervention with a pericardial window have been reported (20).

Takotsubo cardiomyopathy is also known as ampulla cardiomyopathy because of its association with extreme emotional stress. It can also manifest as a complication of severe hypoglycemia in patients with anorexia (7,21–23). Takotsubo cardiomyopathy manifests as reversible left ventricular systolic dysfunction in the absence of coronary artery syndrome and is thought to result from myocardial stunning from a multivessel coronary artery spasm (Movie 1) (22,23). At imaging, Takotsubo cardiomyopathy is characterized by akinesis or ballooning of the apical region of the ventricle, with hypercontraction of the basal segment (7,23). It is named for its resemblance to the Japanese "octopus pot" during contraction, although other less-common patterns have been described (24). Acute mitral regurgitation is seen in approximately 25% of cases (24).

Pulmonary Manifestations.—In patients with severe malnutrition, reductions in elastin, hydroxyproline, surfactant, and antioxidants result in connective tissue damage and emphysema-like structural changes in the lung parenchyma that are similar to those seen in smokers (Fig 5) (7,25). To promote overall survival, lung tissue is thought to be metabolized to provide a substrate for the brain and muscle (26). Bronchiectasis and formation of bullae may also occur (7). Associated complications include a spontaneous pneumothorax, soft-tissue emphysema, and pneumomediastinum, although it is important to distinguish a spontaneous pneumomediastinum because of underlying emphysema from pneumomediastinum secondary to esophageal perforation (Figs 6, E6) (7,26).

Gastrointestinal and Peritoneal Manifestations.—In patients with EDs, ascites can occur because of hypoalbuminemia from malnutrition, liver damage, and resultant portal hypertension

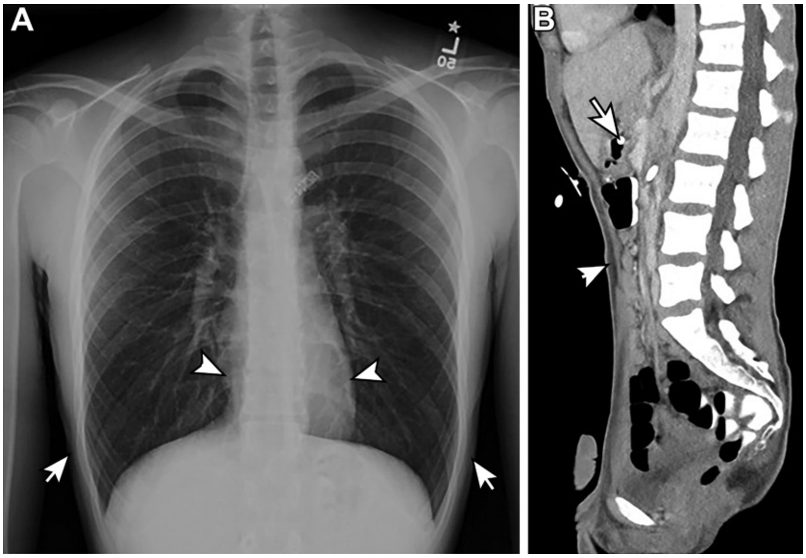


Figure 4. Malnutrition in a 21-year-old man. **(A)** Anteroposterior chest radiograph shows a paucity of subcutaneous soft tissue (arrows) and a small shadow on the heart (arrowheads). **(B)** Sagittal contrast-enhanced CT image of the abdomen and pelvis shows a markedly scaphoid abdomen with little to no subcutaneous and intra-abdominal fat (arrowhead). An enteric tube is partially visible (arrow). Note part of a zipper from an article of the patient's clothing projecting over the left upper chest.

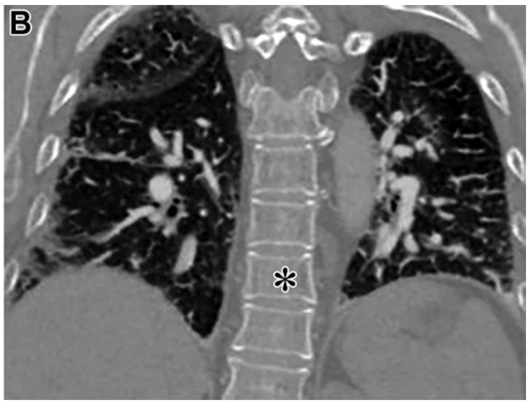


Figure 5. Pulmonary emphysema in a 53-year-old woman with anorexia. **(A)** Axial nonenhanced CT image of the chest shows diffuse pulmonary emphysematous changes (arrows). **(B)** Coronal nonenhanced CT image (bone window) shows diffuse demineralization and trabeculation of the vertebral bodies (*).

or chronic pancreatitis (Fig 2) (27). Ascites in these patients may also be idiopathic; although it is rarely seen in patients with anorexia nervosa without hypoalbuminemia, it has been documented (27).

Musculoskeletal Manifestations.—Nutrient deficiency associated with anorexia in children and adolescents can lead to growth delay, resulting in skeletal immaturity and short stature (26). Growth delay can be evaluated by determining bone age with the use of radiography of the hand or by evaluating apophyseal ossification of the iliac crest (using the Risser classification system) (7) (Fig 7).

Deficiency of vitamin C, which is vital to the synthesis of collagen and acts as an antioxidant, has been documented in a few cases of anorexia nervosa (28,29). A variety of imaging findings have been described (Table 5); however, inflammatory arthropathy may manifest with similar radiographic findings, and findings at MRI are also relatively nonspecific and can be misleading for infectious or noninfectious osteomyelitis, hematologic malignancy, or metastatic disease (29,30). Although findings are often bilateral and symmetric, unilateral manifestation has also been described (29).

Manifestations due to Dehydration and Electrolyte Abnormalities

Central pontine myelinolysis or osmotic demyelination syndrome is a rare neurologic disorder that occurs because of rapid correction of hyponatremia, when dehydration of brain tissue and noninflammatory demyelination of white matter results in symmetric white matter demyelination in the central pons (31,32). Although central pontine myelinolysis is more commonly seen in patients with alcohol overuse and water-electrolyte abnormalities (ie, from polydipsia and/or abuse of diuretics), it is also associated with malnutrition, including that resulting from EDs (31–33). In patients with EDs, central pontine myelinolysis occurs in those with hyponatremia, hypokalemia and metabolic alkalosis, hypogly-

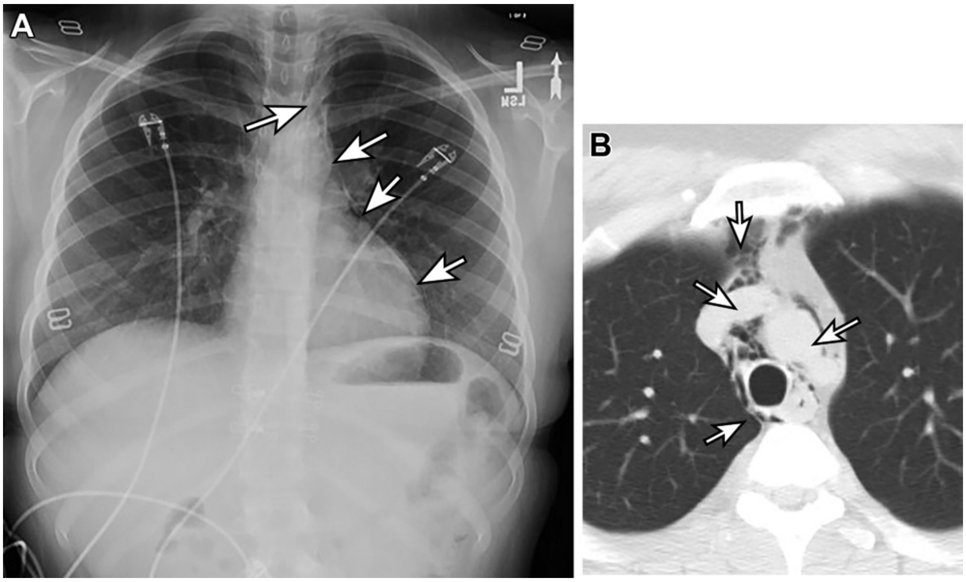


Figure 6. Pneumomediastinum in two patients. **(A)** Anteroposterior chest radiograph in a 14-year-old girl with an eating disorder shows lucencies along the left cardiomediastinal border (arrows). **(B)** Axial nonenhanced CT of the chest (lung window) in a 24-year-old man who presented after an episode of vomiting shows lucencies in the upper mediastinum (arrows).

Figure 7. Delayed skeletal growth in an adolescent boy with anorexia nervosa and a chronological age of 15 years, 9 months. Radiograph of the left hand shows a bone age best resembling that of a boy aged 13 years, 6 months.

Table 5: Radiographic and MRI Findings Associated with Vitamin C Deficiency

Radiography
Osteopenia or cortical thinning
Erosive changes, joint space narrowing
Lucent metaphyseal band (scorbutic zone) underlying dense zone of provisional calcification (Frankel line)
Lucent area on diaphyseal side of Frankel line occurring secondary to disorganized osteoid (Trümmerfeld zone)
Thin sclerotic rim around osteopenic epiphysis (Wimberger ring sign)
Pelkan spur (reflecting healing metaphyseal fractures)
Periosteal reaction secondary to subperiosteal hemorrhage
Soft-tissue swelling
MRI
T1-weighted MRI: diffuse multifocal decreased signal intensity
T2-weighted MRI: increased signal intensity of bone marrow, most frequently in the metaphyses
Periosteal and soft-tissue signal intensity abnormalities, periosteal fluid
Variable bone marrow enhancement

Sources.—References 29 and 30.

cemic coma, and hypophosphatemia (31,33). Central pontine myelinolysis may be asymptomatic or may manifest with encephalopathy, coma, or even death (32). MRI can be used to confirm the diagnosis of central pontine myelinolysis. Diffusion-weighted MRI demonstrates diffusion restriction of the central pons, with sparing of the periphery, which is a finding that is usually seen as early as 1 day from the onset of symptoms (32). T2-weighted and T2-weighted fluid-attenuated inversion-recovery MRI images demonstrate increased signal intensity in a “bat wing"–shaped distribution in the central pons (typically seen later in the disease process) (Fig E7) (32). The area of demyelination enhances at gadolinium-enhanced T1-weighted MRI (33). Because imaging findings may not be seen for up to 2 weeks after the onset of symptoms, the absence of findings does not exclude central pontine myelinolysis. If imaging is negative for these findings and clinical suspicion remains high, MRI should be repeated within 2 weeks (32).

Manifestations due to Hormonal Dysregulation

Genital Manifestations.—Severe malnutrition suppresses the hypothalamic-pituitary axis, decreasing the production of estrogen, which ultimately results in oligomenorrhea or secondary amenorrhea (7). Amenorrhea is considered in the diagnostic criteria for anorexia nervosa and may occur in one-half of patients with bulimia (7). Adolescent girls may present with primary amenorrhea and demonstrate female genital tract immaturity, with a uterus and ovaries that are smaller than expected for the patient's age (26). Specific imaging findings include a uterine body smaller than the cervix; a thin or undetectable endometrium; reduced ovarian volume, with few or absent follicles; and no evidence of cyclic activity in the ovarian tissue (ie, absence of dominant follicles) (Fig 8) (19).

Musculoskeletal Manifestations.—Osteopenia or osteoporosis is common in patients with anorexia nervosa secondary to nutrient deficiency and endocrine derangements that cause estrogen deficiency (7,26). Long-term changes in bone mineralization and architecture are encountered in patients with anorexia, especially when anorexia is diagnosed in adolescence and lasts longer than 5 years. On radiographs, osseous demineralization manifests with cortical thinning and hyperlucent bones (Fig 9). On CT images, cortical thinning and accentuated trabeculae are seen (Figs 5, 9). Osteoporosis is associated with a high risk of fragility or insufficiency fractures, which are best diagnosed with MRI (Fig 10).

In a study of adolescent girls with anorexia nervosa (34) in which dual-energy x-ray absorptiometry and peripheral quantitative CT were used, bone mineral density values were lower than expected for patient age, with the lumbar spine being the most frequently affected area. As such, osteoporosis and multiple fractures occurring at an unexpectedly young age could signify the presence of EDs and should inspire further investigation into the patient's medical history and clinical context.

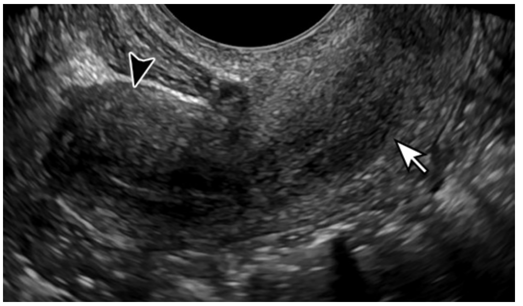


Figure 8. Reproductive immaturity and secondary amenorrhea in a 20-year-old woman with anorexia. Sagittal transvaginal gray-scale US image of the pelvis shows a uterine body (arrowhead) that is approximately the same length as the cervix (arrow), which is indicative of an immature uterus.

Manifestations due to Metabolic Alterations and Changes in Tissue Composition and Distribution

Gastrointestinal Manifestations.—Investigations into the mechanism behind anorexia-induced hepatic steatosis suggest that anorexia-induced lipolysis results in the late accumulation of triglycerides and free fatty acids in the liver and kidneys (35). This finding is also present in patients who abuse alcohol and can suggest it as a comorbidity, particularly in those with bulimia (36). Malnourished patients with EDs may also show pancreatic atrophy, which is more severe in those with a lower body mass index (8). However, this finding improves with nutritional rehabilitation (8).

Soft-Tissue Changes.—Patients with anorexia nervosa demonstrate a low body mass index and a low body fat mass, with a marked deficiency of subcutaneous fat that can be seen at radiography and CT (Fig 11) (26,37). Body fat distribution can be assessed with CT and can be used to indirectly assess tissue quality and composition by measuring tissue attenuation in Hounsfield units (37). Adipose tissue with higher lipid content and larger adipocytes shows lower attenuation, while adipose tissue with lower lipid content, smaller adipocytes, and markers of systemic fibrosis demonstrates higher attenuation (37). Although patients with anorexia nervosa have comparable relative fat distribution compared with normal-weight control subjects, they demonstrate a statistically significant lower fat mass and differences in fat composition (ie, higher fat attenuation) (37).

A complication of severe malnutrition and an associated reduction of retroperitoneal fat is superior mesenteric artery (SMA) syndrome, which occurs when the third part of the duodenum is compressed between the aorta and the SMA at its origin (Fig 12) (8,38). Although this condition is

Figure 9. Osteoporosis and osteoporotic fractures in a 48-year-old woman with anorexia. **(A)** Sagittal nonenhanced CT image of the lumbar spine shows compression fractures of the L1 and L2 vertebral bodies (arrows). **(B)** Anteroposterior radiograph of the right ankle obtained on the same day as **A** shows a distal fibular fracture (arrow). **(C)** Axial nonenhanced CT image of the left foot confirms these fracture findings (arrows).

Figure 10. Stress reaction and fragility fractures in a 45-year-old woman with a history of anorexia nervosa, a body mass index of 17, and osteoporosis. **(A)** Axial T2-weighted fat-saturated MR image of the pelvis shows a nondisplaced fracture of the left inferior pubic ramus, with associated soft-tissue edema (arrows). **(B)** Coronal T2-weighted short tau inversion-recovery MR image of the pelvis shows a stress reaction in the bilateral sacral ala (arrows). **(C)** Coronal T2-weighted short tau inversion-recovery MR image shows patchy bone marrow signal intensity across the bilateral femora (arrows), which is likely caused by hematopoietic marrow.

not specific to restrictive EDs, because it can also occur owing to external compression by a belt or a spica cast, anatomic defects or congenital anomalies, or recent spinal surgery, it can manifest after acute excessive ingestion of food or water in those who already have a paucity of visceral fat (38). An imaging finding that is strongly suggestive of SMA syndrome includes a dilated stomach with a transition point at the third part of the duodenum as it crosses the midline (8). CT criteria for SMA syndrome include an aortic-SMA angle less than 22°–25° in the sagittal plane, or an aorto-mesenteric distance of less than 8–10 mm (Fig 12). SMA syndrome can be complicated by intestinal obstruction, which may be life threatening (8).

Other more subtle and less acutely threatening findings that are associated with decreased

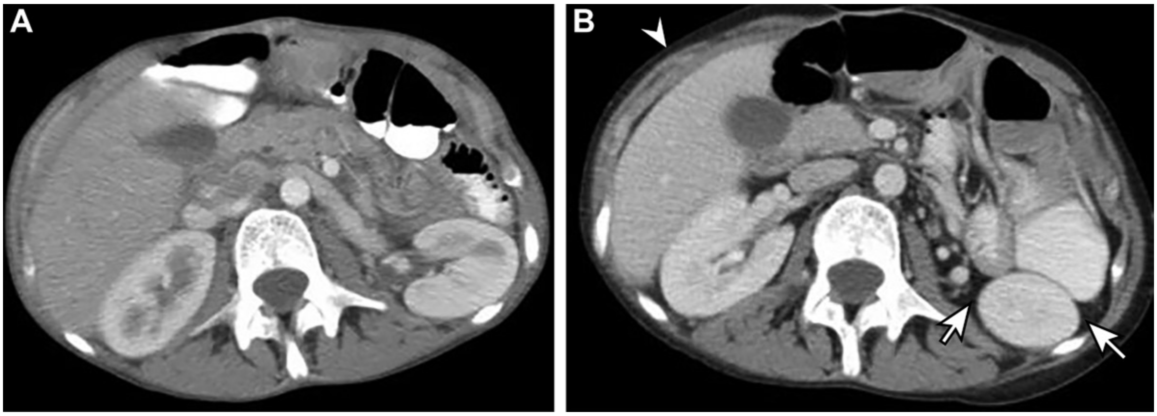
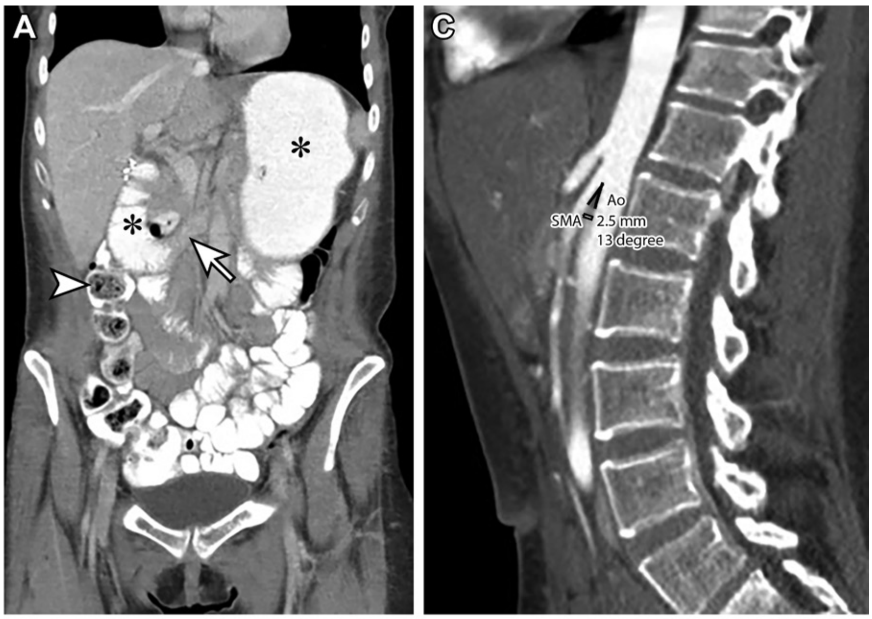


Figure 11. Changes in body fat in a 23-year-old woman with anorexia and bulimia and a body mass index of 12. **(A)** Axial contrast-enhanced CT image of the abdomen and pelvis shows the near absence of subcutaneous, retroperitoneal, and intraperitoneal fat. **(B)** Axial contrast-enhanced CT image of the abdomen and pelvis acquired 1 year later shows a mild increase in intra-abdominal (arrows) and subcutaneous (arrowhead) body fat.



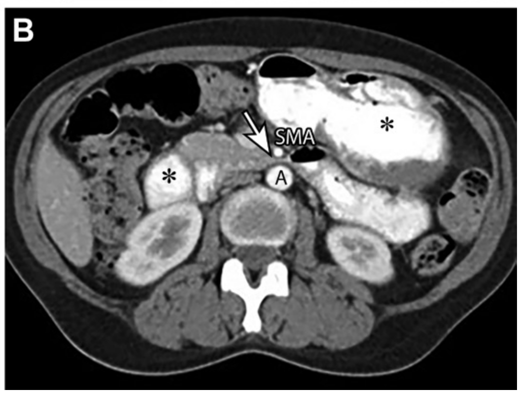


Figure 12. Superior mesenteric artery (SMA) syndrome in a 28-year-old woman with anorexia nervosa who lost 17 lb (7.7 kg) in 2 months and had multiple episodes of abdominal pain after excessive eating. **(A, B)** Contrast-enhanced coronal **(A)** and axial **(B)** CT images of the abdomen and pelvis show a stomach and first portion of the duodenum distended with oral contrast material and ingested material (*), although oral contrast material is also noted to fill several loops of bowel (arrowhead in **A**). Note the sharp narrowing of the second and third portions of the duodenum (arrow) as it courses between the SMA and the aorta *(A)*. The duodenum is partially collapsed as it passes between the vessels. **(C)** Sagittal CT angiogram of the abdomen and pelvis reveals a scaphoid abdomen and a narrow angle of SMA origin from the aorta, with an aorto-mesenteric distance of only 2.4 mm and an aorto-SMA angle of 13.5°. *Ao* = abdominal aorta.

overall body fat mass include a patulous eustachian tube and enophthalmos. A patulous eustachian tube may be seen when there is reduction in the fatty tissue that normally keeps the eustachian tube closed (6). As a result, the tube intermittently stays open. At CT, this manifests with the abnormal presence of air throughout most of the eustachian tube (Fig 13). Related findings that may be seen include dilatation of the eustachian tube orifices, widening of the nasopharyngeal recesses, or flattening of the posterior wall of the pharynx (6). In a similar manner to the development of a patulous eustachian tube, enophthalmos involves the loss of intraorbital fat, leading to progressive posterior displacement of the globe in the orbit. At imaging, this is suggested by greater than 50% coverage of the globe (Fig 14).

Musculoskeletal Manifestations.—Bone marrow adipose tissue (BMAT) has become a more recent focus of research and may be a biomarker

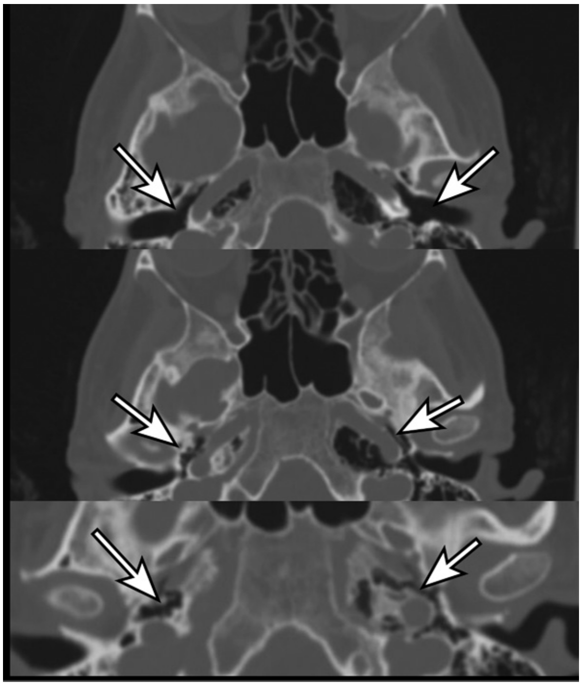


Figure 13. Patulous eustachian tube in a 36-year-old man with anorexia nervosa. Axial nonenhanced CT images of the head show dilatation of the eustachian orifices and proximal tubes (arrows).

Figure 14. Enophthalmos in an 18-year-old woman with anorexia nervosa. Axial T2-weighted MRI of the brain shows a small amount of subcutaneous fat over the orbits. The measurements show greater than 50% coverage of the globe.

for osteoporosis (39). BMAT can be assessed with dual-energy CT or MRI (39). The metabolism of BMAT during late-stage starvation results in gelatinous transformation, known as serous atrophy of bone marrow, in which adipocytes and hematopoietic cells decrease in size and number and the marrow is replaced by hyaluronic acid–rich mucopolysaccharides (39). In patients with anorexia nervosa, BMAT is higher in the lumbar spine and femur, with a statistically significant negative correlation between BMAT and bone mineral density (39). Serous atrophy of bone marrow appears as diffuse signal hypointensity at T1-weighted MRI and diffusely increased signal intensity at fat-suppressed T2-weighted and short-tau inversion-recovery MRI (39). Changes in signal intensity can also be seen in areas of subcutaneous and visceral adipose tissue (39). It is important to note that changes in signal intensity may be misinterpreted as technical errors from unsuccessful fat suppression (39). Proton MR spectroscopy is currently the reference standard for in vivo quantification of BMAT, particularly point-resolved spectroscopy (PRESS) and stimulated echo acquisition mode (STEAM) sequences (39). Dual-energy CT shows promising results for evaluation of both bone mineral density and BMAT in the lumbar spine and can be used to evaluate these simultaneously. The ability to account for BMAT, which can confound assessment of bone mineral density, allows improved quantification of bone mineral density (39). Dual-energy CT also correlates well with hydrogen 1 (^{1}H) MR spectroscopy for quantification of BMAT (39).

Purging Disorders

Repetitive Vomiting: Manifestations of Physical Insult or Prolonged Irritation

***Dental Manifestations.*—**In patients who purge, repetitive vomiting exposes the dentition to acidic vomitus, which, over time, leads to erosions (6). The lingual surfaces of the anterior maxillary teeth tend to be most affected, followed by the lingual and occlusal surfaces of the upper molars (6). Substantial dental work in these teeth combined with additional imaging findings seen in patients with EDs should increase confidence in the diagnosis.

***Gastrointestinal Manifestations.*—**Although repetitive vomiting is classically associated with some EDs (ie, bulimia nervosa), patients with EDs, in general, can present with symptoms of dyspepsia and vomiting (8). Repetitive self-induced vomiting results in injury to the esophageal mucosa, which increases the risk of esophageal inflammation and esophageal mucosal damage (8). This can manifest as esophageal wall thickening, erosive changes, and formation of ulcers that can be seen at imaging (26). Chronic inflammation can lead to esophagitis and strictures (Fig 15; Movie 2A, 2B) (6,26). Although there are limited data, researchers suspect that repeated exposure to acid can lead to Barrett esophagus. Given the association of Barrett esophagus with a 30-times increased risk of esophageal carcinoma, patients with EDs are theoretically at increased risk for malignancy (8,26). Cases of patients with bulimia who presented with worsening epigastric pain and reflux and were found to have esophageal adenocarcinoma have been described (8). Limited data also suggest that patients who are hospitalized for EDs have a six-times increased risk of developing esophageal squamous cell carcinoma, although tobacco use, alcohol overuse, and chronic nutritional deficiency are confounding factors (8).

Repetitive vomiting is associated with increased laxity of the lower esophageal sphincter, which may manifest as a patulous gastroesophageal junction at fluoroscopy and can even result in esophageal rupture (ie, Boerhaave syndrome) if vomiting is aggressive (Figs 16, 17) (6,7). Boerhaave syndrome is associated with a high mortality rate without early diagnosis and surgical repair, and it can be indicated by the presence of pneumomediastinal or pleural effusions at chest radiography or chest CT, mediastinal fluid or inflammation at chest CT, or a leak at fluoroscopy (6). Similar to the esophagus, the gastric wall can become inflamed and thickened, leading to gastritis and formation of ulcers as a result of repetitive binging and purging (Fig 18) (26).

Pulmonary Manifestations.—Repetitive vomiting increases the risk of aspiration, but diagnosis can be difficult because patients with EDs tend to withhold an accurate history (26). Sudden-onset respiratory distress and imaging that shows focal or multifocal opacities in the lower lobe or dependent portions of the lungs in an otherwise healthy young patient should raise concern for self-induced vomiting (Fig 19). At CT, opacities may have a ground-glass, tree-in-bud, or consolidative appearance, depending on the degree or severity of aspiration or on the nature of the aspirated material; segmental or lobar atelectasis may also occur. Although opacities from simple aspiration typically resolve within 24–48 hours, persistent or worsening opacities beyond 3 days suggest the presence of a superimposed infection or development of acute respiratory distress syndrome. Aspiration can also be complicated by formation of an empyema or abscess (Fig 20).

Manifestations due to Nervous System Dysregulation

Parotid gland swelling, or sialadenosis, is seen in 10%–25% of patients with bulimia nervosa and occurs 2–6 days after cessation of purging behaviors (6). It is thought to occur secondary to a combination of abnormal sympathetic innervation, chronic autonomic stimulation, and malnutrition. Sialadenosis may be unilateral or bilateral and tends to be painless (6). Although most cases resolve spontaneously, progression to persistent sialadenomegaly has been observed in approximately 0.5% of cases (6). At nonenhanced CT, sialadenosis is associated with increased attenuation of the parotid gland secondary to decreased intraparotid fat. At contrast-enhanced CT, increased parotid enhancement is secondary to glandular hypertrophy and increased vascularity (Fig 21) (7).

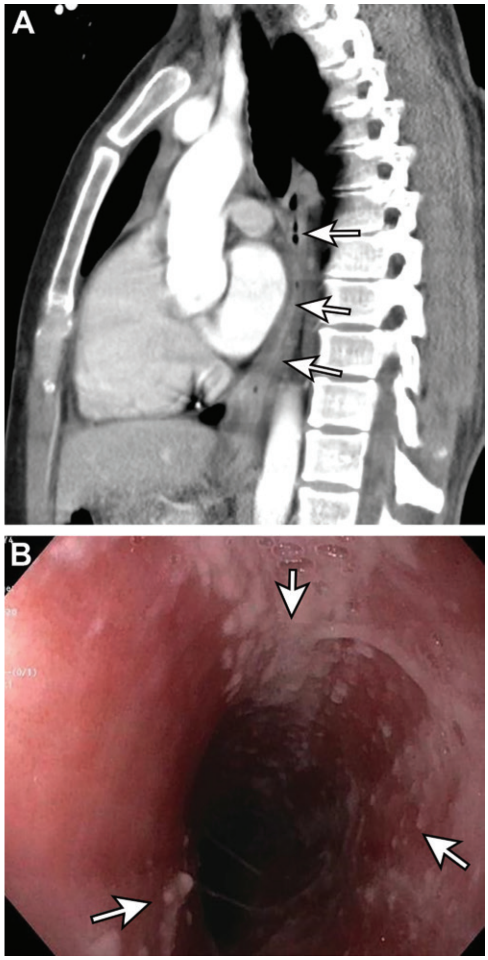


Figure 15. Esophagitis in a 31-year-old woman with bulimia. **(A)** Sagittal contrast-enhanced CT image of the chest shows wall thickening and luminal narrowing of the distal esophagus (arrows). **(B)** Image from upper gastrointestinal endoscopy shows severe esophagitis (arrows).

Laxative Abuse

Pulmonary Manifestations.—In patients who regularly use mineral oil as a laxative, chronic aspiration can result in exogenous lipoid pneumonia (40). Findings can be unilateral or bilateral and range from diffuse consolidations to reticular or nodular patterns of opacities. The lower and middle lobes are most commonly affected. At thin-sec-

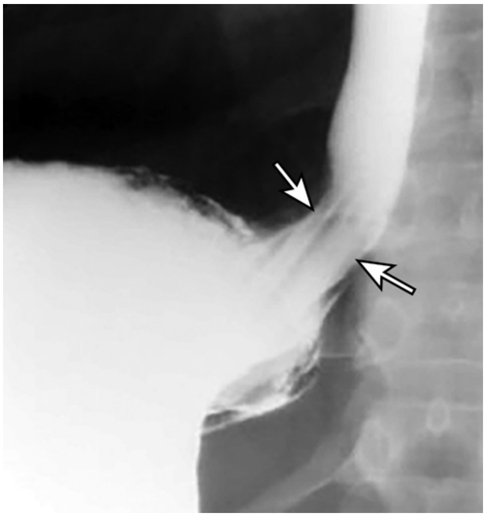


Figure 16. Patulous gastroesophageal junction in a 34-year-old woman with bulimia. Esophagram shows direct spillage of oral contrast material from the esophagus into the stomach lumen through a dilated gastroesophageal junction (arrows). Note the apparent extension of rugal folds above the diaphragm, suggesting the presence of a hiatal hernia.

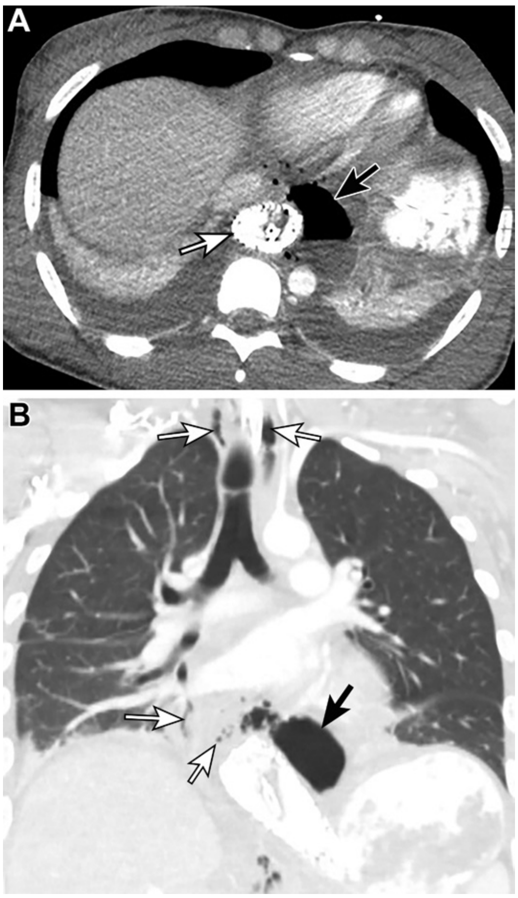


Figure 17. Esophageal rupture in a 28-year-old woman after attempts at self-induced vomiting with repeated dry heaving. Axial **(A)** and coronal **(B)** contrast-enhanced CT images of the chest obtained after placement of an esophageal stent show a rounded lucency (black arrow), which was originally seen on a chest radiograph (not shown), and additional foci of air (white arrows).

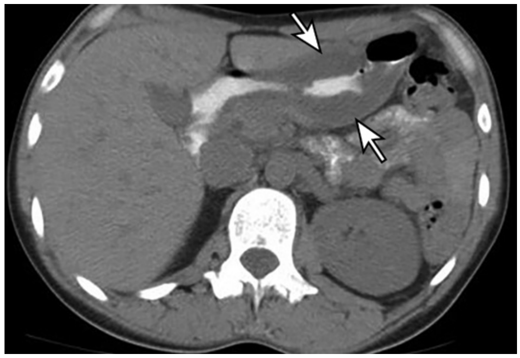


Figure 18. Gastric inflammation in a 37-year-old woman with bulimia nervosa. Axial nonenhanced CT image shows severe wall thickening of the distal stomach (arrows).

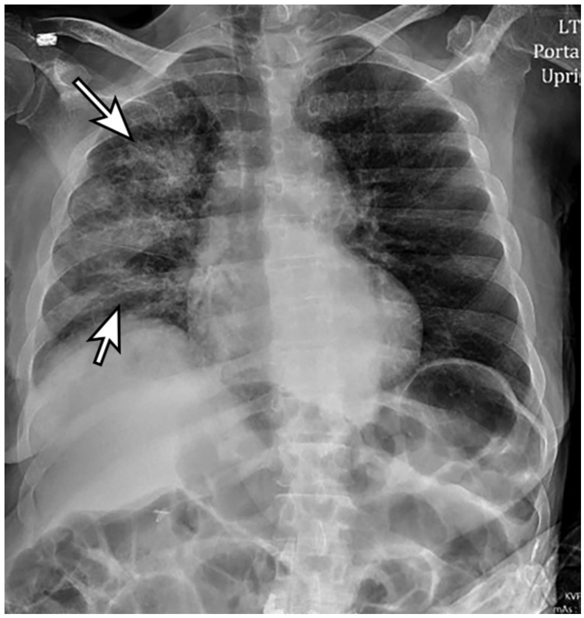


Figure 19. Recurrent aspiration pneumonia in a 52-year-old woman with anorexia nervosa and depression who presented with shortness of breath after multiple bouts of vomiting. Chest radiograph shows diffuse patchy opacities in the right lung (arrows).

tion CT, consolidations, ground-glass opacities, a "crazy paving" pattern, interlobular septal thickening, or masslike lesions can be seen, with consolidations with negative attenuation values (−30 to −150 HU) suggesting the presence of fat (Fig 22). Correct windowing at CT improves identification of fat and facilitates accurate diagnosis.

Gastrointestinal Manifestations.—Cathartic colon refers to colonic changes from chronic use of stimulant laxatives, a method used by approximately 60% of patients with EDs (7). In patients who chronically abuse stimulant laxatives, constipation develops because of a combination of progressive dependence on the laxatives to drive peristalsis and increased medication tolerance

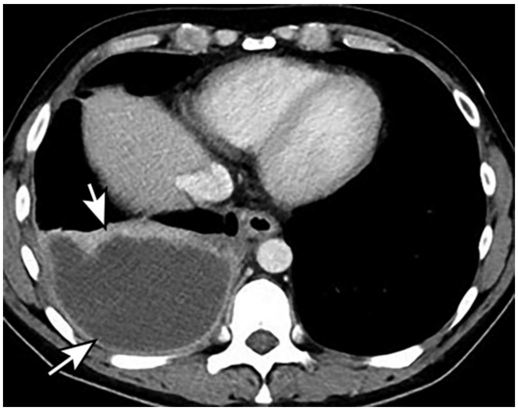


Figure 20. Empyema in a 41-year-old woman with anorexia nervosa and bipolar disorder who presented after 3 months of vomiting and pain in the right upper quadrant of the abdomen. Axial contrast-enhanced CT image of the abdomen shows a complex fluid collection with an enhancing rim in the right hemothorax, which is indicative of empyema (arrows). Nonenhanced CT of the chest (not shown) performed on a later day allowed assessment of the entire collection and adjacent lung.

Figure 21. Chronic enlargement of the parotid gland in a 67-year-old woman with bipolar disorder and vomiting. Coronal contrast-enhanced CT image of the neck shows bilateral enlarged hyperattenuating parotid glands (*). Note that the normal parotid glands show mainly the same attenuation as that of fat and are not enlarged.

(41). This can be associated with melanosis coli, which is brown pigmentation of the colon seen at colonoscopy or pathologic examination (7). Symptoms of cathartic colon are nonspecific, including abdominal pain, diarrhea, and weight loss (7). At cross-sectional imaging, severe cases may be associated with nonspecific colonic wall thickening, loss of haustra, and increased submucosal fat (Fig 23) (7,26). Administration of a barium enema before imaging may reveal a loss of colonic haustra and transverse folds, colonic dilatation, a patulous ileocecal valve, dilatation of the terminal ileum, or pseudostrictures. The ascending colon is mainly affected (42). Because the loss of haustra in the cathartic colon can resemble chronic ulcerative colitis, care should be taken to correlate imaging findings with clinical history and laboratory testing results.

Manifestations due to Hormonal Dysregulation

In patients who purge, upregulation of aldosterone persists even after termination of purging behaviors, leading to development of lower extremity edema from continued sodium and water retention (6). Peripheral edema also occurs in approximately 20% of patients with anorexia nervosa (2).

Binging and Other Forms of Disordered Ingestion

Gastrointestinal Manifestations

Disorders associated with binging, including bulimia nervosa, can result in chronic distention of

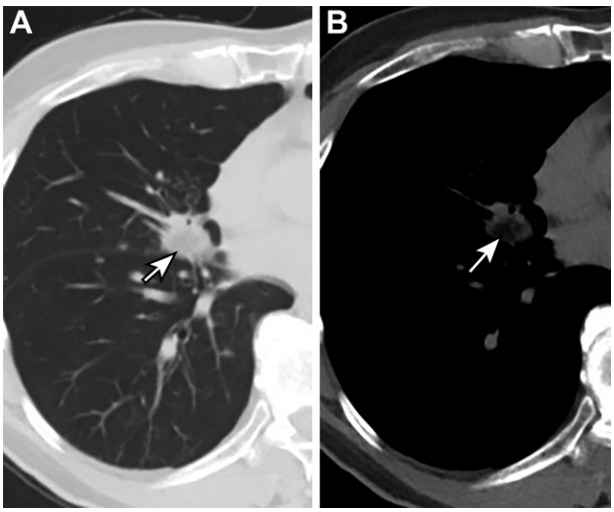


Figure 22. Lipoid pneumonia in a 74-year-old man who had taken mineral oil nightly for 32 years. **(A)** Axial nonenhanced CT image of the chest (lung window) shows a spiculated lung nodule in the left lower lobe (arrow). **(B)** Axial nonenhanced CT image of the chest, with adjusting windowing, shows a fat component in the nodule, with surrounding ground-glass opacities (arrow).

the stomach, with an increase in stomach capacity to more than 15 L (43). Normally, rupture can occur with intragastric pressures greater than 120 mm Hg (approximately 4 L of fluid) (43). Binging episodes can lead to acute gastric dilatation, which can have potentially life-threatening complications including gastric necrosis, perforation or rupture, and even abdominal compartment syndrome (Fig 24) (43–45).

Foreign Bodies and Bezoars.—The term *pica* refers to the consumption of nonfood substances

Figure 23. Cathartic colon in a 31-year-old woman with bulimia nervosa. Coronal contrast-enhanced CT image shows diffuse colonic edema with minimal associated mesenteric stranding. There is loss of colonic haustration (arrows). The findings are indicative of cathartic or laxative abuse. Hepatic steatosis is also present.

Figure 24. Acute gastric dilatation in a 19-year-old woman after a binge-eating episode. Upright abdominal radiograph shows a markedly dilated stomach (*) filled with ingested materials. A large amount of stool is seen in the ascending colon (arrow).

without nutritional value (7,46). This is most commonly seen in children, patients with learning disabilities, and pregnant women (7). There are several subtypes of pica (ie, *trichophagia*, the consumption of hair; and *geophagia*, the consumption of dirt) (7,46). Imaging findings suggestive of pica include the presence of foreign bodies in the gastrointestinal tract (Figs 25, E8, E9). This can include the formation of bezoars, which are condensed masses of undigested or partially digested matter (Figs 26–28). In patients with trichophagia, trichobezoars may form and are suggested by their mottled appearance at radiography and CT (Figs 26, 27) (46). Foreign body ingestion and pica can be complicated by poisoning, obstruction, or perforation, for which surgical intervention may be required (Figs 27, 29) (46).

Various foods consumed in normal amounts under normal circumstances can have an unusual appearance at imaging and potentially be mistaken for foreign bodies; clinical correlation of imaging with the patient's recent meals, medical history, and other imaging findings (or lack thereof) can help prevent a misdiagnosis (Fig E10, Movies 3, 4).

Comorbidities and Mimics

Substance Use Disorder

Compared with 9% of the general population, approximately 50% of patients with EDs also abuse alcohol or other illicit substances (36). For example, in a 2015 study, Fouladi et al (36) found that patients with bulimia nervosa demonstrated the highest rates of alcohol consumption in comparison with patients with other EDs. EDs and substance use disorders share several common psychoemotional and behavioral features, including impulsivity and problems with loss of control, with tendencies toward self-destructive behavior (36). Several substances, including tobacco, cocaine, and amphetamines, have the perceived added "benefits" of suppressing the appetite and increasing energy.

EDs and substance use disorders have multiple common imaging manifestations (47). EDs and alcohol overuse can both result in hepatic steatosis, pancreatitis, cerebral volume loss, and Wernicke encephalopathy. Just as with EDs, tobacco use can be associated with emphysema, pneumothorax, or pneumomediastinum. Myocardial infarction and aspiration pneumonia can be seen in patients with EDs and in those who abuse cocaine. As with EDs, the use of amphetamines can result in the development of cardiomyopathy. Use of ipecac to induce vomiting can have cardiac consequences, including cardiomyopathy and congestive heart failure (6).

Medically Induced Anorexia

Medical conditions can lead to the development of imaging findings that are similar to those in

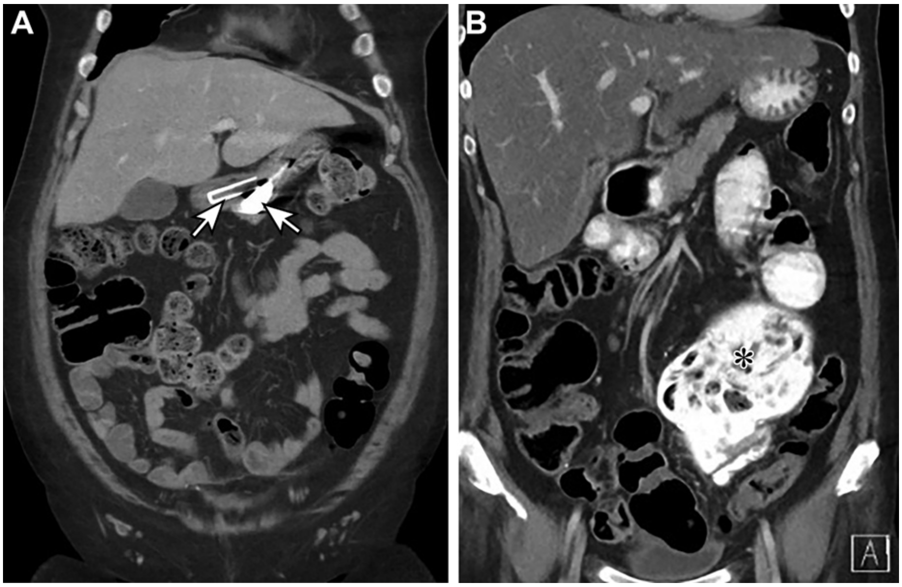


Figure 25. Pica in a 45-year-old woman with multiple foreign body ingestions, multiple emergency room visits, and multiple prior surgical procedures. **(A)** Coronal contrast-enhanced CT image of the abdomen and pelvis shows multiple foreign bodies in the stomach (arrows), which were confirmed to represent multiple ingested batteries. **(B)** Coronal contrast-enhanced CT image obtained at a different visit shows a distended portion of the small bowel containing a bezoar (*). Note the diffuse hypoattenuation of the liver, which is indicative of fatty liver changes.

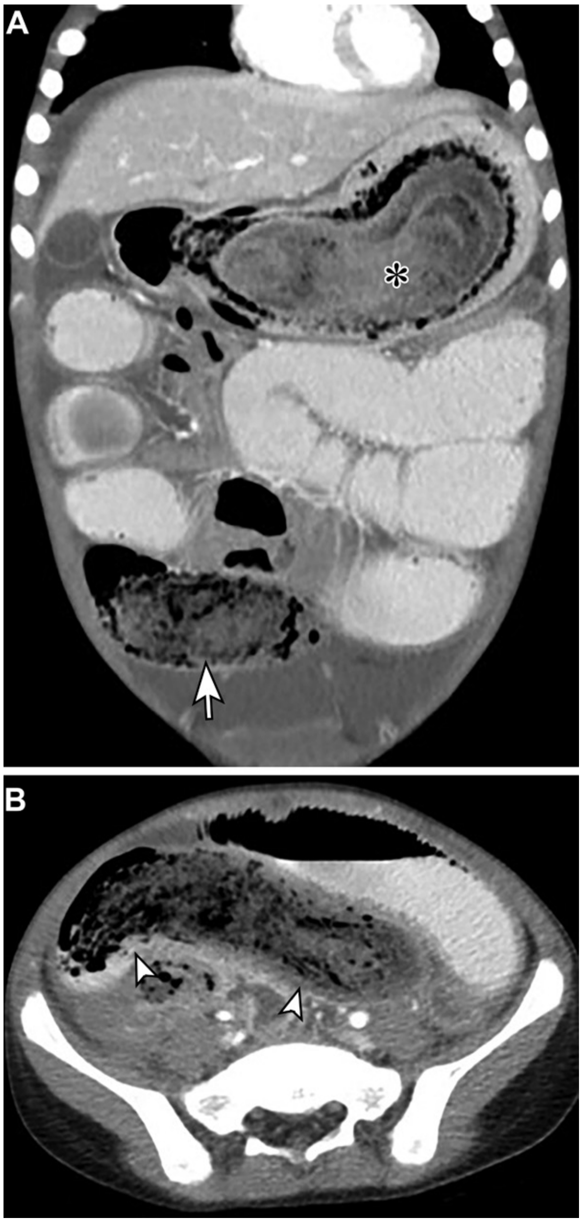


Figure 26. Trichobezoars in a 6-year-old girl with trichotillomania. Coronal **(A)** and axial **(B)** CT images of the abdomen and pelvis show a large bezoar in the stomach (* in **A**), a second large obstructing bezoar in the proximal ileum (arrow in **A**), and a third bezoar in the more distal ileum (arrowheads in **B**).

patients with EDs. Cancer-induced cachexia, a paraneoplastic syndrome, is characterized by poor oral intake and metabolic disturbances (48,49). Nausea and vomiting are common in patients with advanced cancer or those receiving chemotherapy and can limit nutrient intake (48). A syndrome of autonomic insufficiency that is associated with abnormalities in gastric emptying contributes to nausea and early satiety (Fig E2) (48). Cancer-induced cachexia can also result from central nervous system involvement (48). In addition, patients with head or neck and esophageal carcinomas can experience dysphagia or odynophagia, and pancreatic insufficiency can lead to malabsorption in patients with pancreatic tumors (48). Other factors contributing to malnutrition in these patients include decreased taste sensation, pain, and depression (48). Radiation therapy, chemotherapy, and surgery increase nutrient requirements but can damage the gastrointestinal tract, impairing nutrient absorption (48). Infections in immunosuppressed patients (eg, patients with cancer), including those with tuberculosis or HIV infection, can also result in cachexia (50).

Child Abuse or Neglect

Failure to thrive describes lack of normal growth and is characterized by a child whose growth is persistently below the 3rd–5th percentile on standard growth chart (51). Although older children may demonstrate a lack of growth due to mal-

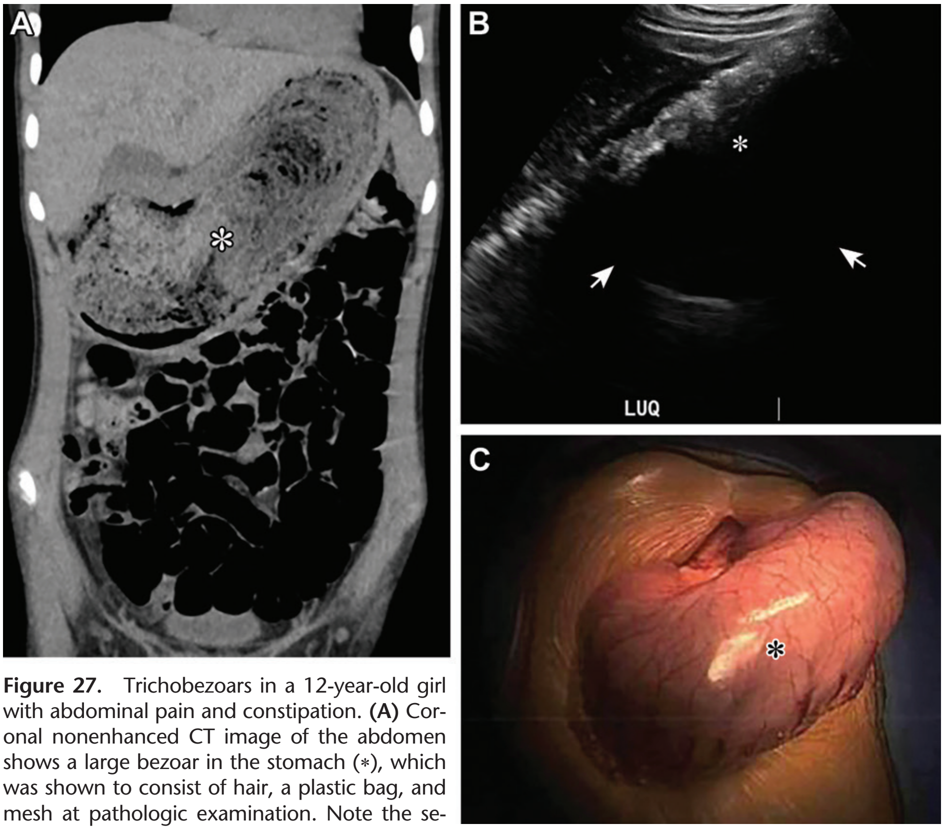


Figure 27. Trichobezoars in a 12-year-old girl with abdominal pain and constipation. **(A)** Coronal nonenhanced CT image of the abdomen shows a large bezoar in the stomach (*), which was shown to consist of hair, a plastic bag, and mesh at pathologic examination. Note the severe paucity subcutaneous fat. **(B)** Transverse gray-scale US image of the left upper quadrant of the abdomen shows strong posterior shadowing produced by the bezoar (*) and contents (arrows). **(C)** Intraoperative photograph demonstrates a markedly distended hyperemic stomach filled with a bezoar (*) and ingested material .

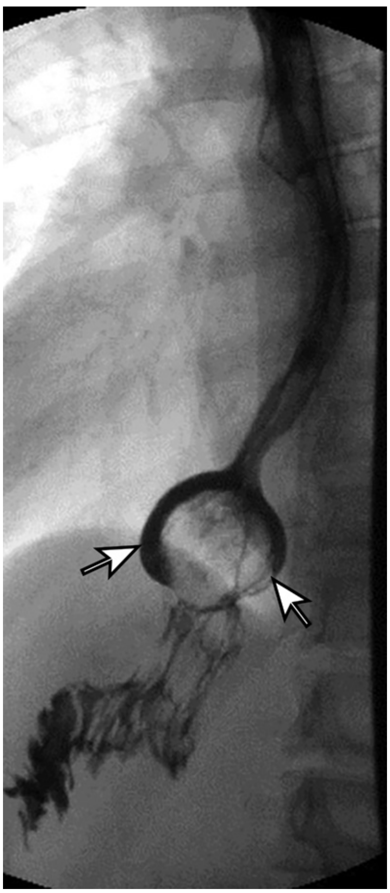


Figure 28. Trichobezoars in a 21-year-old woman with dysphasia and a history of multiple prior surgical procedures for ingestion of foreign bodies. Upper gastrointestinal endoscopic image of the esophagus shows a round filling defect at the level of the gastrojejunal pouch (arrows) indicative of a bezoar.

nutrition from neglect, fatal starvation can occur in younger infants and toddlers who cannot feed themselves (51). Imaging manifestations of neglect include a depressed fontanel, decreased subcutaneous and visceral fat, enophthalmos, thymic involution, muscle atrophy, osteopenia or osteoporosis, rickets, an empty gastrointestinal tract, and hepatic steatosis (51,52).

Abuse or Neglect of Older Adult Patients

Millions of older adults experience abuse or neglect, and numbers are expected to increase with increasing longevity (53). In evaluating older patients, it is important not to dismiss potentially concerning findings as natural changes of aging: older patients develop osteoporosis and are prone to fragility fractures, particularly of the hips, proximal humeri, and vertebral bodies. Rib fractures are common after a fall or resuscitation. However, fractures in atypical locations

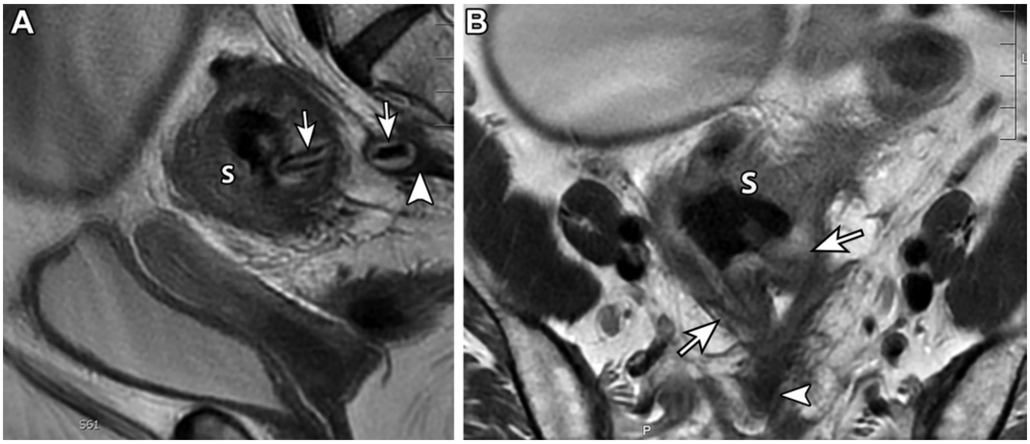


Figure 29. Perforation of the sigmoid colon secondary to ingestion of a foreign body in a 61-year-old woman with schizoaffective disorder and pica. Sagittal **(A)** and axial **(B)** T2-weighted MR images of the pelvis show a foreign body with linear limbs (arrows), which was presumed to be a nail clipper, protruding from the posterior wall of the distal sigmoid colon (*S*) with surrounding soft-tissue thickening and inflammatory changes (arrowhead). Given that the patient had multiple medical comorbidities and that the foreign body was nonobstructive, surgery was deferred. Note the right adnexal cystic mass.

may suggest abuse, and untreated fractures may suggest medical neglect. Neglect of older patients may also be suggested by the presence of severe muscle atrophy, soft-tissue infections, decubitus ulcers, and osteomyelitis (53).

EDs in the Age of COVID-19

COVID-19 and the associated lockdown period have had a substantial psychopathologic effect on patients with EDs for a number of reasons (54). This period has increased multiple stressors. Increased use of social media has resulted in heightened self-awareness and body insecurity, home confinement and forced cohabitation have led to increased interpersonal and family conflicts, and the media's highlighting of threats of food shortages has led to food insecurity. For example, studies have demonstrated that lockdown and the shift from in-person programs to telehealth have interfered with the recovery process for patients with bulimia nervosa by creating conditions that are counterproductive to treatment, resulting in exacerbation of binge eating (representing a possible symptom of emotional dysregulation) and compensatory physical activity. Similarly, although patients with anorexia have demonstrated progressive weight gain, they have also demonstrated exacerbation of compensatory physical exercise (54).

Conclusion

EDs have a variety of multisystemic manifestations that can be seen at imaging, although some can be quite subtle. Although EDs are common and are associated with high morbidity and mortality rates, they tend to be underdiagnosed and undertreated. Because of this, identification of potential sequelae at imaging can help to suggest or establish the diagnosis.

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